

LIRA System Use Cases

Final Document

1. Manage Library Catalog

Scenario	A librarian updates the core repository of library materials.									
Triggering Event	Librarian initiates an add, edit, or delete action in the catalog dashboard.									
Brief Description	Central administrative function for maintaining library records synchronized with Koha ILS.									
Actors	Librarian (Primary), Koha (Secondary)									
Preconditions	Librarian is logged in with administrative privileges.									
Postconditions	Catalog database is updated and synchronized with Koha ILS. The AI Processing Service indexes the record.									
Flow of Activities	<table><tr><th>Actor</th><th>System</th></tr><tr><td>1. Librarian logs in to the LIRA admin portal with administrative credentials.</td><td>1.1 System validates credentials and redirects to the admin dashboard.</td></tr><tr><td>2. Librarian selects 'Add New Resource,' 'Edit,' or 'Delete'.</td><td>2.1 System displays the catalog metadata form.</td></tr><tr><td>3. Librarian inputs or updates metadata and submits.</td><td>3.1 System validates metadata. 3.2 System commits record to Koha ILS. 3.3 AI Processing Service indexes it. 3.4 System displays confirmation.</td></tr></table>		Actor	System	1. Librarian logs in to the LIRA admin portal with administrative credentials.	1.1 System validates credentials and redirects to the admin dashboard.	2. Librarian selects 'Add New Resource,' 'Edit,' or 'Delete'.	2.1 System displays the catalog metadata form.	3. Librarian inputs or updates metadata and submits.	3.1 System validates metadata. 3.2 System commits record to Koha ILS. 3.3 AI Processing Service indexes it. 3.4 System displays confirmation.
Actor	System									
1. Librarian logs in to the LIRA admin portal with administrative credentials.	1.1 System validates credentials and redirects to the admin dashboard.									
2. Librarian selects 'Add New Resource,' 'Edit,' or 'Delete'.	2.1 System displays the catalog metadata form.									
3. Librarian inputs or updates metadata and submits.	3.1 System validates metadata. 3.2 System commits record to Koha ILS. 3.3 AI Processing Service indexes it. 3.4 System displays confirmation.									
Exception Conditions	- 1.1 If authentication fails, system denies access. - 3.1 If Koha ILS synchronization fails or metadata is invalid, system saves changes locally in a queue and alerts the librarian.									

2. Manage Book Loans

Scenario	Librarian processes the borrowing or returning of a physical book.											
Actors	Librarian (Primary), Koha (Secondary)											
Preconditions	Patron has a valid library account with no active blocks.											
Flow of Activities	<table><tr><th>Actor</th><th>System</th></tr><tr><td>1. Patron presents a book and ID.</td><td>1.1 System is ready to scan.</td></tr><tr><td>2. Librarian scans ID and barcode.</td><td>2.1 System retrieves patron record. 2.2 System verifies no active blocks. 2.3 System updates status to 'Borrowed'.</td></tr><tr><td>3. Librarian confirms due date.</td><td>3.1 System monitors loan duration.</td></tr><tr><td>4. (System-Initiated) Cron job evaluates dates.</td><td>4.1 System triggers automated overdue notification if due date passes.</td></tr></table>		Actor	System	1. Patron presents a book and ID.	1.1 System is ready to scan.	2. Librarian scans ID and barcode.	2.1 System retrieves patron record. 2.2 System verifies no active blocks. 2.3 System updates status to 'Borrowed'.	3. Librarian confirms due date.	3.1 System monitors loan duration.	4. (System-Initiated) Cron job evaluates dates.	4.1 System triggers automated overdue notification if due date passes.
Actor	System											
1. Patron presents a book and ID.	1.1 System is ready to scan.											
2. Librarian scans ID and barcode.	2.1 System retrieves patron record. 2.2 System verifies no active blocks. 2.3 System updates status to 'Borrowed'.											
3. Librarian confirms due date.	3.1 System monitors loan duration.											
4. (System-Initiated) Cron job evaluates dates.	4.1 System triggers automated overdue notification if due date passes.											
Exception Conditions	- 2.2 If a patron has reached their maximum borrowing limit, system rejects transaction.											

3. Manage Chatbot

Scenario	Librarian configures the AI chatbot's behavior and maintains its FAQ.													
Actors	Librarian (Primary), AI (Secondary)													
Flow of Activities	<table><tr><th>Actor</th><th>System</th></tr><tr><td>1. Librarian logs in to Chatbot Config.</td><td>1.1 System displays current settings.</td></tr><tr><td>2. Librarian adjusts parameters and FAQ.</td><td>2.1 System validates settings.</td></tr><tr><td>3. Librarian deploys to sandbox.</td><td>3.1 System applies settings to sandbox.</td></tr><tr><td>4. Librarian tests configuration.</td><td>4.1 System generates responses based on sandbox.</td></tr><tr><td>5. Librarian deploys to production.</td><td>5.1 System updates production parameters.</td></tr></table>		Actor	System	1. Librarian logs in to Chatbot Config.	1.1 System displays current settings.	2. Librarian adjusts parameters and FAQ.	2.1 System validates settings.	3. Librarian deploys to sandbox.	3.1 System applies settings to sandbox.	4. Librarian tests configuration.	4.1 System generates responses based on sandbox.	5. Librarian deploys to production.	5.1 System updates production parameters.
Actor	System													
1. Librarian logs in to Chatbot Config.	1.1 System displays current settings.													
2. Librarian adjusts parameters and FAQ.	2.1 System validates settings.													
3. Librarian deploys to sandbox.	3.1 System applies settings to sandbox.													
4. Librarian tests configuration.	4.1 System generates responses based on sandbox.													
5. Librarian deploys to production.	5.1 System updates production parameters.													
Exception Conditions	- 2.1 If parameter config is invalid, system prevents deployment. - 2.1 If there is insufficient chat data for FAQ, system prompts for manual FAQ creation.													

4. Manage Reports

Scenario	Librarian requests operational analytics on library usage.									
Actors	Librarian (Primary), Koha (Secondary)									
Flow of Activities	<table><tr><th>Actor</th><th>System</th></tr><tr><td>1. Librarian selects metrics and timeframe.</td><td>1.1 System validates parameters. 1.2 System queries database. 1.3 System compiles data.</td></tr><tr><td>2. Librarian clicks 'Generate Report'.</td><td>2.1 System displays report and export options.</td></tr><tr><td>3. Librarian selects export format.</td><td>3.1 System triggers file download.</td></tr></table>		Actor	System	1. Librarian selects metrics and timeframe.	1.1 System validates parameters. 1.2 System queries database. 1.3 System compiles data.	2. Librarian clicks 'Generate Report'.	2.1 System displays report and export options.	3. Librarian selects export format.	3.1 System triggers file download.
Actor	System									
1. Librarian selects metrics and timeframe.	1.1 System validates parameters. 1.2 System queries database. 1.3 System compiles data.									
2. Librarian clicks 'Generate Report'.	2.1 System displays report and export options.									
3. Librarian selects export format.	3.1 System triggers file download.									
Exception Conditions	- 1.2 If requested date range contains no data, system displays an empty-state message.									

5. View Library Catalog

Scenario	A patron browses or searches the library's collection.							
Actors	Student/Faculty (Primary), Koha (Secondary)							
Flow of Activities	<table><tr><th>Actor</th><th>System</th></tr><tr><td>1. User enters keywords or filters.</td><td>1.1 System queries Koha ILS. 1.2 System displays matching items.</td></tr><tr><td>2. User selects an item.</td><td>2.1 System renders complete catalog record.</td></tr></table>		Actor	System	1. User enters keywords or filters.	1.1 System queries Koha ILS. 1.2 System displays matching items.	2. User selects an item.	2.1 System renders complete catalog record.
Actor	System							
1. User enters keywords or filters.	1.1 System queries Koha ILS. 1.2 System displays matching items.							
2. User selects an item.	2.1 System renders complete catalog record.							
Exception Conditions	- 1.2 If no matching records are found, system displays 'No materials found' and suggests broader terms.							

6. Borrow Books & Track Active Loans

Scenario	A patron requests to borrow an available library resource, and views their dashboard to track current borrowings and deadlines.																	
Actors	Student/Faculty (Primary), Koha (Secondary)																	
Flow of Activities	<table><tr><th>Actor</th><th>System</th></tr><tr><td colspan="2">Part A: Borrow Book Transaction</td></tr><tr><td>1. User clicks 'Borrow' on an available book.</td><td>1.1 System validates user's borrowing status.</td></tr><tr><td>2. User confirms borrow details.</td><td>2.1 System records loan request in Koha. 2.2 System updates item availability to Pending/Borrowed.</td></tr><tr><td>3. (No Action)</td><td>3.1 System displays confirmation message.</td></tr><tr><td colspan="2">Part B: Track Active Loans</td></tr><tr><td>4. Patron accesses dashboard.</td><td>4.1 System retrieves active loans from Koha ILS. 4.2 System displays list of borrowed items & due dates.</td></tr><tr><td>5. (No Action)</td><td>5.1 System checks if any items are nearing due dates and highlights them.</td></tr></table>		Actor	System	Part A: Borrow Book Transaction		1. User clicks 'Borrow' on an available book.	1.1 System validates user's borrowing status.	2. User confirms borrow details.	2.1 System records loan request in Koha. 2.2 System updates item availability to Pending/Borrowed.	3. (No Action)	3.1 System displays confirmation message.	Part B: Track Active Loans		4. Patron accesses dashboard.	4.1 System retrieves active loans from Koha ILS. 4.2 System displays list of borrowed items & due dates.	5. (No Action)	5.1 System checks if any items are nearing due dates and highlights them.
Actor	System																	
Part A: Borrow Book Transaction																		
1. User clicks 'Borrow' on an available book.	1.1 System validates user's borrowing status.																	
2. User confirms borrow details.	2.1 System records loan request in Koha. 2.2 System updates item availability to Pending/Borrowed.																	
3. (No Action)	3.1 System displays confirmation message.																	
Part B: Track Active Loans																		
4. Patron accesses dashboard.	4.1 System retrieves active loans from Koha ILS. 4.2 System displays list of borrowed items & due dates.																	
5. (No Action)	5.1 System checks if any items are nearing due dates and highlights them.																	
Exception Conditions	- 1.1 If user reached borrowing limit, system denies request. - 2.1 If item is no longer available, system notifies user and suggests similar items. - 4.1 If Koha ILS is unreachable, system displays error message.																	

7. Ask AI Chatbot

Scenario	A patron queries the AI chatbot for assistance, summaries, or citations.							
Actors	Student/Faculty (Primary), AI (Secondary)							
Flow of Activities	<table><tr><th>Actor</th><th>System</th></tr><tr><td>1. User types question or requests action.</td><td>1.1 System routes query to AI and shows typing indicator. 1.2 System analyzes intent and extracts metadata.</td></tr><tr><td>2. User reviews generated response.</td><td>2.1 System generates answer, summary, or citation. 2.2 System updates user profile to refine future results.</td></tr></table>		Actor	System	1. User types question or requests action.	1.1 System routes query to AI and shows typing indicator. 1.2 System analyzes intent and extracts metadata.	2. User reviews generated response.	2.1 System generates answer, summary, or citation. 2.2 System updates user profile to refine future results.
Actor	System							
1. User types question or requests action.	1.1 System routes query to AI and shows typing indicator. 1.2 System analyzes intent and extracts metadata.							
2. User reviews generated response.	2.1 System generates answer, summary, or citation. 2.2 System updates user profile to refine future results.							
Exception Conditions	- 1.1 If AI Service times out, system displays graceful fallback message. - 1.2 If query is inappropriate, system generates courteous refusal. - 1.2 If source metadata is incomplete for citation, system prompts user to manually input missing fields.							